

AUTOMATIC FAN CONTROLLER

For An Active Heatsink

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This circuit is based on a thermistor and an operational amplifier (op-amp) that drives a MOSFET to switch a cooling fan on and off in an active heatsink system. The design also includes an overload protection mechanism.

An active heatsink is normally included in electronic products such as power supplies and power amplifiers. It has a fan attached to it to actively pull heat away from it and the chip underneath. The automatic fan controller constantly monitors heatsink temperature and switches on the fan whenever the temperature rises above a pre-determined level. A typical active heatsink used in a class D power amplifier is shown in Fig. 1.



Fig. 1: A typical active heatsink in an audio power amplifier (Credit: www.parts-express.com)

You may think that a fan controller is not required and that the fan should always be on. While that may be true to some extent, in order to reduce the irritating fan noise and prevent undesirable power consumption when power dissipation in the system is normal, the controller accomplishes the goal quite satisfactorily.

Circuit and working

The circuit diagram of the automatic fan controller for an active heatsink is shown in Fig. 2. It is built around op-amp LM358 (IC1), two MOSFETs BS170 (T1 and T2), a small 12V cooling fan and a few other components.

As you can see from the circuit diagram, LM358 (IC1) has two op-amps. The first has pins 2 and 3, and its output is pin 1. The second op-amp has pins 5 and 6, and its output is pin 7.

The first op-amp forms a simple comparator between the voltage set by 10k NTC thermistor TH1 and 15-kilo-ohm resistor R1. The fan on threshold level (fan_set) is determined by potentiometer VR1, and hysteresis is controlled by positive-feedback resistor R2.

The thermistor is placed on the heatsink of the chip to be protected,

PARTS LIST	
Semiconductors:	
IC1	- LM358 op-amp
T1, T2	- BS170 MOSFET
D1	- 1N4007 rectifier diode
LED1	- 5mm LED
TH1	- 10-kilo-ohm NTC thermistor
Resistors (all 1/4-watt, $\pm 5\%$ carbon):	
R1	- 15-kilo-ohm
R2	- 47-kilo-ohm
R3	- 150-kilo-ohm
R4	- 2.2-kilo-ohm
VR1	- 10-kilo-ohm potmeter
VR2	- 22-kilo-ohm potmeter
Capacitors:	
C1	- 100 μ F, 25V electrolytic
C2, C3	- 0.1 μ F ceramic disk
Miscellaneous:	
CON1	- 2-pin terminal connector
CON2, CON3	- 2-pin connector
	- 12V DC, 350mA cooling fan
	- 12V regulated power supply
	- Heatsink for cooling fan

as shown in Fig. 3. Resistance of the thermistor decreases as temperature rises.

When temperature exceeds the set threshold level, output of the first op-amp goes high. This turns on BS170 (T1) to start the connected 12V (< 350mA) cooling fan.

While a thermally-controlled heatsink fan is generally a good design, a thermal overload protection for the main system is also required.

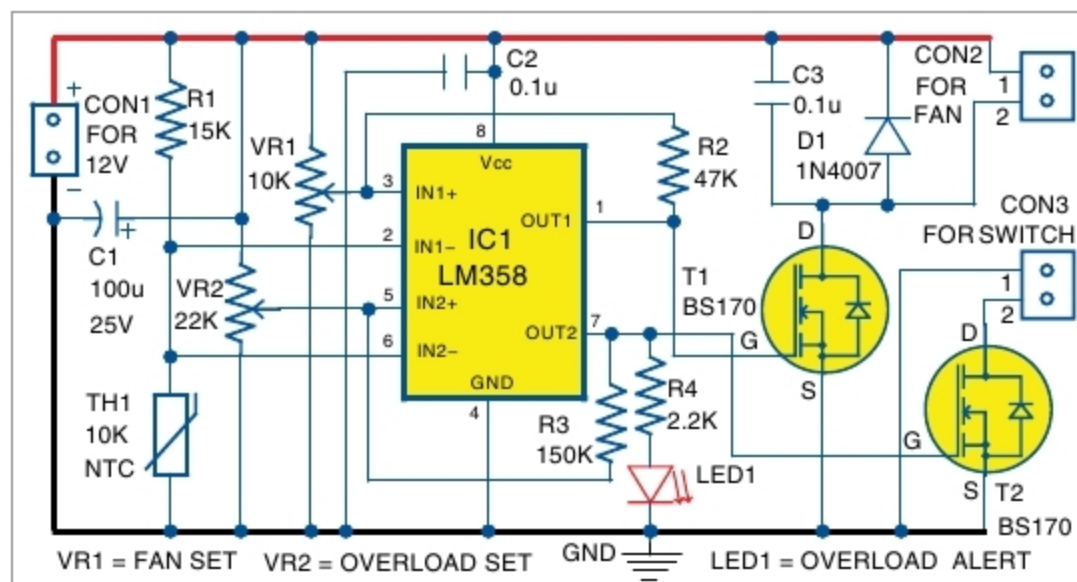


Fig. 2: Circuit diagram of automatic fan controller for active heatsink

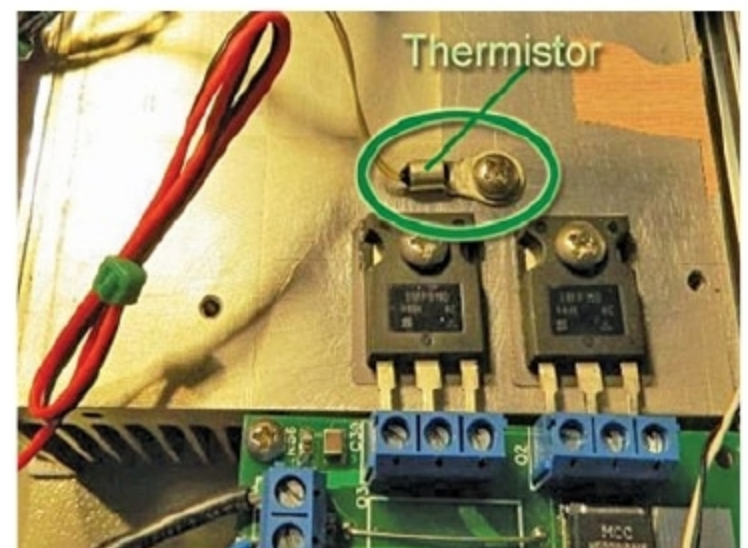


Fig. 3: Thermistor mounted on heatsink assembly of power supply unit